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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0329– MOBILE COMPUTING

PROJECT TITLE

"1G to 5G Evolution Interactive Learning Platform."

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ABSTRACT

- The 1G to 5G Evolution Interactive Learning Platform is a mobile app that helps students learn how mobile networks changed from 1G to 5G.
- It shows each generation's data rate, spectrum band, latency, technologies and services in a simple, visual way.

Purpose of the Project

- Learners explore an interactive timeline, compare generations using charts, and take gamified quizzes.
- The app records quiz score, engagement time and comparison accuracy to measure learning.
- It makes a difficult topic easy, interesting and available anytime on a phone.

INTRODUCTION

- Mobile networks have grown from 1G (voice only) to 5G (very high speed, low delay and smart devices).
- Textbooks explain this evolution in long text and tables, which students find hard to remember.
- This app turns the same content into an interactive timeline, charts and quizzes.

Importances :

- This system
  - makes learning simple and interesting
  - helps compare generations easily
  - tests understanding through quizzes

PROBLEM STATEMENTS AND OBJECTIVES

Problem Statement:

- Students find it difficult to remember mobile network generations from 1G to 5G.
- Comparing speed, frequency, latency, and services is confusing.
- Traditional study materials do not provide interactive learning or instant feedback.

Objectives

Content and Visualization Module

- To display 1G to 5G information using timelines and charts.
- To make comparing network generations easy.

Quiz and Assessment Module

- To conduct interactive quizzes and provide instant feedback.
- To track quiz scores, learning time, and comparison accuracy.

EXISTING SYSTEM AND PROPOSED SYSTEMS

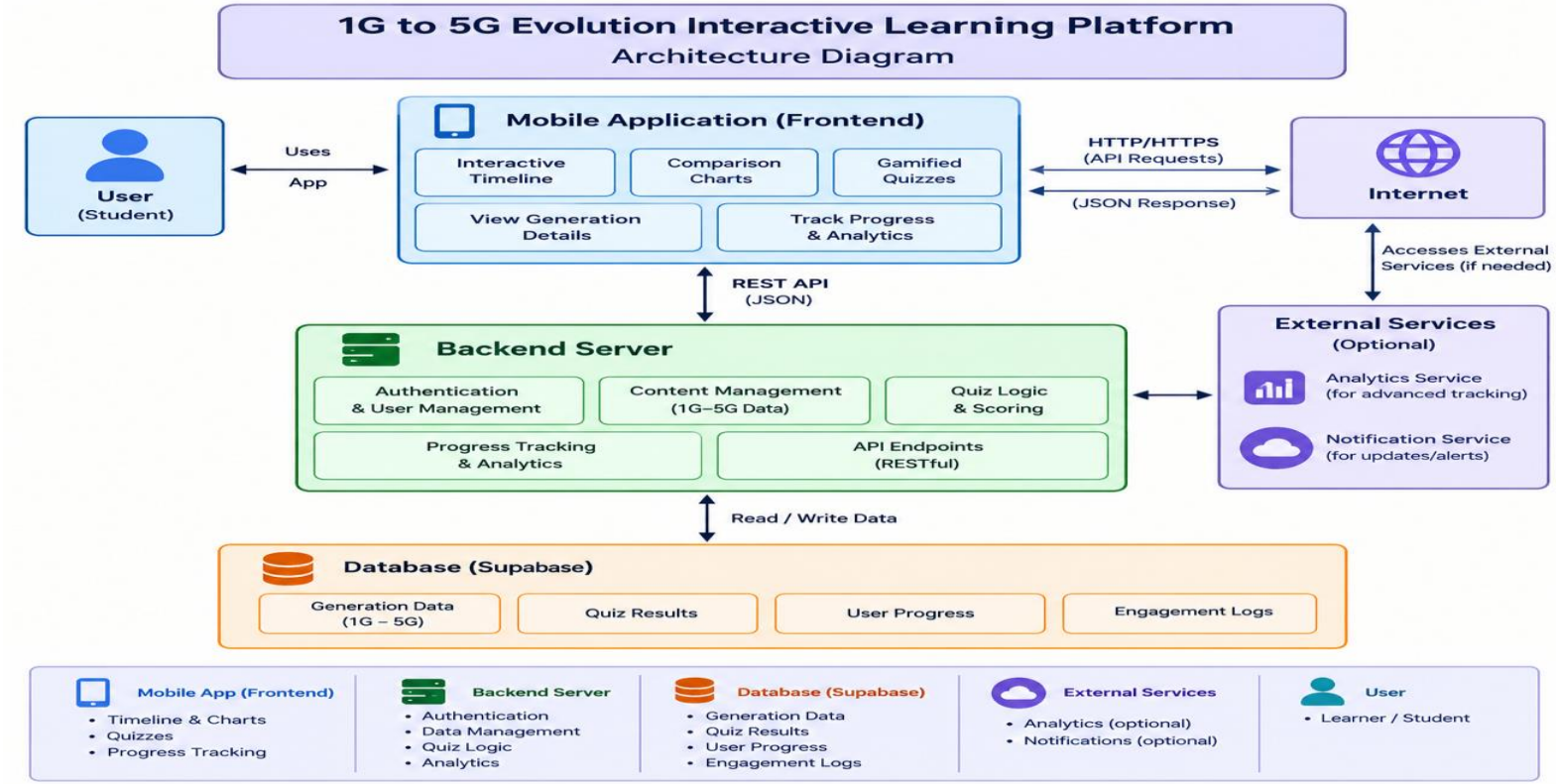
Existing System:

- Students learn from textbooks, notes and slides.
- No interaction or practice for the learner.
- Comparing generations takes a lot of effort.
- No tracking of learner progress.

Proposed System:

- An interactive Android / React Native app for 1G to 5G.
- Timeline shows the evolution step by step.
- Charts compare data rate, latency and spectrum band.
- Gamified quizzes give feedback; progress is tracked.

METHODOLOGY



Module -1: Interactive Timeline & Data Visualisation

- This module shows the evolution from 1G to 5G on a scrollable timeline.
- The learner taps a generation to see its data rate, spectrum band, latency, technologies and services.
- Bar and line charts (MPAndroidChart) compare generations side by side.
- Data comes from the SQLite content store, so it works offline.

Key Functions:

- Displays each generation in order of time.
- Shows details of the selected generation.
- Draws charts to compare two or more generations.

MODULE 2: Gamified Quiz & Progress Tracking

- This module tests what the learner has understood.
- Questions come from SQLite, with points, levels, badges and instant feedback.
- It saves quiz score, engagement time and comparison accuracy for each learner.
- Feedback from a student group is used to improve the app.

Key Functions:

- Picks questions by generation or topic.
- Calculates the score and gives instant feedback.
- Shows a progress dashboard for the learner.

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IEEE STANDARDS

1. Learning Technology Standards

- IEEE 1484.12.1 – Learning Object Metadata for organising learning content.
- IEEE 1484.1 – Architecture for learning technology systems.

2. Wireless Communication Standards

- IEEE 802.11 – Wi-Fi for downloading and updating content.
- IEEE 802.15.1 – Bluetooth for sharing with nearby devices.

3. Software Engineering Standards

- IEEE 12207 – Software life cycle processes.
- IEEE 29148 – Requirements engineering.

SDG CONTRIBUTION

The project helps students learn telecom concepts in an easy, interactive and affordable way.

- **SDG 4 – Quality Education**
  - Interactive lessons and quizzes improve understanding and support self-paced learning.
- **SDG 9 – Industry, Innovation, and Infrastructure**
  - Builds awareness of 5G and modern communication technology, supporting innovation.
- **SDG 10 – Reduced Inequalities**
  - Offline mobile access lets more students learn, even with limited internet.

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CONCLUSION

- Makes the 1G to 5G evolution simple and easy to understand.
- Interactive timeline and charts help learners compare generations quickly.
- Gamified quizzes improve interest, practice and memory.
- Quiz score, engagement time and comparison accuracy show how well learners progress.
- Works offline on a mobile phone using a SQLite content store.
- Combines education, technology and innovation in a single app.

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Thank You